

TEST AND MEASUREMENT For AUTOMOTIVE ELECTRONICS

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To ensure comfort and safety, the automobile industry needs proper T&M instruments

Automobile manufacturers must ensure customer comfort, well-being and safety. It cannot do that without proper test and measurement (T&M) instruments. Internal processes need to be extremely precise and controlled, because even a minor error can have adverse consequences.

Testing is the process by which automotive companies ensure quality, safety and reliability of automotive systems. This challenge is exponentially getting difficult, as the vehicles can now drive themselves.

To ensure that autonomous vehicles are safer than human drivers, these need extensive testing to help with the design of vehicle components and to validate that the hardware and software perform correctly to increase test coverage, while reducing the cost of tests. With modular and flexible tools, you can future-proof the systems and adapt to new technologies.

Testing power consumption

A multi-channel power probe determines the power consumption of electronic control units (ECUs) in both standby and wake-up modes. It reads voltage and current, and

sends data to the oscilloscope, where it is synchronised with digital bus signals and displayed.

It is possible to measure very small quiescent currents in μA or nA range, as well as high current peaks up to ampere range. With 1MHz bandwidth and 5Msample/s sampling rate, the power probe can even acquire very short current or voltage pulses.

Pawan Kumar, area manager - product support and applications, Rohde-Schwarz (R&S), says, "RT-ZVC probe can be used to add up to 16 additional test channels to RTE or RTO oscilloscopes. R&S offers test solutions for automotive Ethernet that are both comprehensive and compact. RTO oscilloscopes for 100Base-T1/BroadR-Reach allow transmission rates up to 1Gbps."

Testing infotainment systems

An infotainment system delivers entertainment and informational content. Kumar says, "We provide solutions for testing of infotainment systems, like advanced driver assistance system (ADAS), car audio, vehicle-to-everything (V2X), navigation, protocol decoding and eCall.

"An ADAS wide-band chirp signal generation and analysis can be done using

SMW200A and FSW up to 85GHz. 24GHz and 77GHz radars of 2Ghz bandwidth can be easily tested with the test setup.

"A radio frequency (RF) vector signal generator with various radio standards is used for audio-testing AM, FM, DAB, DRM, DRM+, SXM and so on. It is also used for signal generation for GPS, GLONASS, Galileo, Beidou and QZSS. WLAN standard IEEE 802.11p (11p, pWLAN) specifies a technology for V2X communications. We offer

Fig. 1: Test setup for automotive infotainment RF characterisation (Credit: R&S)

